Unit-1 D C Circuit

→ Circuit: The conducting path through which electric current flows.

→ Parameters: The various elements of the electric circuit are called Parameters. eg Resistors, capacitors, Inductor.

→ Linear Circuit: The circuit which are made from resistors, capacitors, Inductors are called linear circuit.

→ Non-linear circuit: The circuit which made up of transistors, diode, transformers are called non-linear circuit.

→ Unilateral circuits: A unilateral is one whose properties or characteristics changes with the direction of operations
eg. diode, rectifiers

→ Bilateral Circuit: It is that circuit whose properties or characteristics are same in either direction.
eg. Transmission line, resistors.

→ Active element: The elements which produce energy in the form of voltage and current. eg. - Transistors, generator.

→ Passive element: These are the element which store energy in the form of voltage and current eg. RLC

→ Charge: The amount of energy an electron has is charge.
$Q = ne$, where $e = 1.6 \times 10^{-19} C$.

→ Electric Current: The rate of flow of charge is called electric current. $I = \frac{Q}{t} = C/sec = A$.

→ Voltage / Electric Potential : Work done in moving a charge to one place to another / The ability to do work.

$$V = \frac{\text{work done}}{Q} = \frac{J}{C} = \text{Volts.}$$

→ Power : The rate at which work is done.

$$V \cdot I = \frac{\text{Work done}}{Q} \cdot \frac{Q}{t} = \frac{\text{work done}}{\text{time}} = \frac{J}{Sec}$$

$\boxed{\text{or } P = I^2R \text{ or } P = \frac{V^2}{R}}$

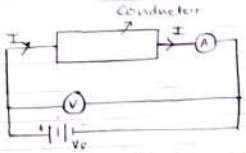
→ Energy : The amount for which the work is done

$$P \cdot t = \frac{J}{Sec} \cdot Sec = J \quad \text{or} \quad Kwhr$$

Types of energy :-

- Heat energy : Also known as Thermal Energy. The energy is transfer from one body to another as the result of difference in temperature.
- Kinetic energy : ~~The~~ Kinetic energy is that in which the energy associated with the movement of object. The K.E. of an object depends on mass and velocity
- Potential energy : The energy ~~which~~ that is stored within the object due to the position of the object.
- Mechanical energy : The sum of kinetic and potential energy in an object that is used to particular work.
- Electrical energy : The energy generated by the movement of electrons from one point to another.
- Photonic energy : The energy carried by a single photon
- Sound energy : When a force such as sound and pressure cause a substance or object to vibrate
- Hydro energy : The energy that is taken from water and converted to electricity

→ Ohm's Law :-



$I \propto V$

$\boxed{V = IR}$

$R = V/I$

It states that the current flowing through a conductor is directly proportional to the potential difference across the ends of the conductor providing physical condition like temp, pressure do not change.

→ Resistance : The property of a substance which opposes the flow of current through it. unit - ohm Ω $\left[\frac{R}{\text{unit}}\right]$

Laws of Resistance :

1) The Resistance is directly proportional to the length of the conductor. $R \propto l$.
2) Resistance are inversely proportional to Area of cross section of conductor $R \propto 1/a$.

$$R \propto \frac{l}{a} \rightarrow \boxed{R = \rho \frac{l}{a}} \rightarrow \boxed{\rho = \frac{Ra}{l}}$$

Resistivity : ρ is the resistivity, Proportionality constant unit - ohm meter (Ωm)

Q. The specific resistance of a platinum is $10.5\ \mu\Omega$. What should be the length of platinum wire to have a resistance of $4\ \Omega$ if diameter is ~~0.74~~ 0.0274 cm.

Sol^n → ~~$R = 10.5 \times 10^{-6}$~~ diameter = 0.0274 cm

~~radius~~ = 2.74×10^{-4} m.

$\rho = 10.5 \times 10^{-8}\ \Omega m$

$R = 4\ \Omega$

$$l = \frac{RA}{\rho} = \frac{4 \times 3.14 \times 2.74 \times 10^{-4} \times 2.74 \times 10^{-4}}{4 \times 10.5 \times 10^{-8}} = 2.24\,m$$

Q. A coil consists of 2000 turns of cu wire having a cross-sectional area of 0.8 mm. The mean ~~length per~~ length per turn is 80 cm and resistivity = 0.02 μΩm find the resistance.

Sol^n $$R = \rho \frac{l}{A} = \frac{0.02 \times 10^{-6} \times 80 \times 2000 \times 10}{0.8 \times 10^{-6} \times 100 \times 100} = 40\,\Omega$$

Q. The heater element is made up of nichrome wire of having $\rho = 100 \times 10^{-8}\ \Omega m$. The diameter of wire is 0.4 mm. Calculate the length to get the resistance 40 Ω.

Sol^n $R = \rho \frac{l}{A}$ → ~~100 × 10~~ $l = \frac{RA}{\rho}$

$$l = \frac{40 \times 3.14 \times 0.2 \times 0.2 \times 10^{-6}}{100 \times 10^{-8}} \Rightarrow l = 5.024\,m$$

Q. The resistance of a conductor 1 mm² in cross section and 20 m long is 0.346 Ω. Determine the specific resistance of the material.

Sol^n $R = \rho \frac{l}{A} = \rho \cdot \frac{RA}{l}$

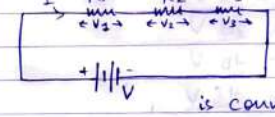
$$= \frac{0.346 \times 1 \times 10^{-6}}{20} = 1.73 \times 10^{-8}\ \Omega m$$

→ Inductors: An inductor also called a coil, choke, or reactor is a passive two terminal electrical component that store energy in a magnetic field

when electrical current flows through. denoted by 'L', unit = Henry (H)

→ Capacitor: It is a device that stores electrical energy in the form of electrical field (electrostatic). It is a passive component with two terminal. The effect of capacitor is known as capacitance. unit = Farad (F).

→ Series and Parallel Circuits:



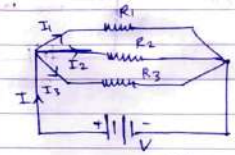
Series arrangement is that type of arrangement ~~in~~ in which two or more the two resistance is connected end to ~~ed~~ end.

$V = IR$,

$V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$

$V = V_1 + V_2 + V_3$ → $V = IR_1 + IR_2 + IR_3$

→ $V = I(R_1 + R_2 + R_3)$, $\boxed{R = (R_1 + R_2 + R_3 ..)}$



The parallel arrangement of circuit is where all the resistance is connected parallelly.

$V = IR$, $I = V/R$

$I_1 = V/R_1$, $I_2 = V/R_2$, $I_3 = V/R_3$

$I = I_1 + I_2 + I_3$ → $\frac{V}{R} = V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$

→ $\boxed{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$

23/8/24

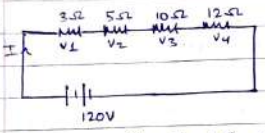
Q. four coils having resistance of 3 Ω, 5 Ω, 10 Ω, 12 Ω and connected in series across 120 V determine:

i) equivalent resistance

ii) current flowing within the resistance

iii) voltage drop across each resistance

Ans.



Given: $R_1 = 3\Omega$, $R_2 = 5\Omega$
$R_3 = 10\Omega$, $R_4 = 12\Omega$.
$V = 120V$

(i) $R_{eq.} = R_1 + R_2 + R_3 + R_4 \Rightarrow [3+5+10+12]\Omega = \underline{30\Omega}$

(ii) Acc. to Ohm's law:

$V_s = I\ R_{eq.}$

$120 = I\ 30$

$I = \underline{4A}$

(iii) $V_1 = IR_1 = 4 \times 3 = \underline{12\ V}$

$V_2 = IR_2 = 4 \times 5 = \underline{20\ V}$

$V_3 = IR_3 = 4 \times 10 = \underline{40\ V}$

$V_4 = IR_4 = 4 \times 12 = \underline{48\ V}$ Ans.

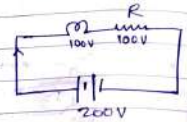
Q. A 100 volt ~~watt~~ 60 watt ~~per~~ bulb is to be operated from 200V supply. What is the resistance to be connected in series with the bulb.

Ans. Bulb: $V = 100\ V$, $P = 60\ W$

$V_s = 200V$.

$P = VI \Rightarrow 60 = 100\ I$

$I = \underline{0.6\ A}$



$R = V/I = \frac{100}{0.6} = \underline{166.67\ \Omega}$

Q. 3 loads A, B, C are connected in parallel to a ~~20~~ 240V source. load A takes 9.6 KW, Load B takes 60A. and load C has a resistance 4.8Ω. Calculate:

(i) R_A & R_B

(ii) Total Current

(iii) Total Power

(iv) Equivalent resistance.

Ans. (i) $R_A \Rightarrow 9.6\ KW \rightarrow 9600W$

$P = VI \Rightarrow 240 \times I = 9600$

$I = \frac{9600}{240} = 40A$

$R = \frac{V}{I} = \frac{240}{40} = \underline{6\Omega}$.

$R_B \Rightarrow V = IR \Rightarrow 240 = 60 \times R$

$R = \frac{V}{I} = \frac{240}{60} = \underline{4\Omega}$

(ii) Total Current: $I = I_A + I_B + I_C$

$= 40 + 60 + 50$

$= \underline{150A}$

$I_C = \frac{V_s}{R_C} = \frac{240}{4.8} = \underline{50A}$

(iii) Total Power: $P = VI$

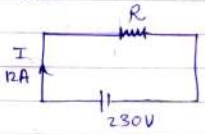
$P = 240 \times 150 = \underline{3600\ Watt}$ or $\underline{3.6\ kW}$

(iv) Equ. Resistance: $\frac{1}{R_{eq.}} = \frac{1}{R_A} + \frac{1}{R_B} + \frac{1}{R_C} \Rightarrow \frac{1}{6} + \frac{1}{4} + \frac{1}{4.8}$

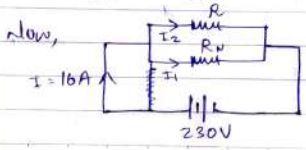
$\frac{1}{R_{eq.}} = \frac{30}{48} \Rightarrow R_{eq.} = \frac{48}{30} = \underline{\frac{8}{5}\ \Omega}$

Q. When a resistance is place across 230V supply the current is 12A. What is the value of resistor that must be connected in parallel to increase the load current to 16A.

Ans. $R = \frac{V}{I} = \frac{230}{12} = 19.1\ \Omega$.



Now,



$I = I_1 + I_2$

$16 = I_1 + 12 \Rightarrow I_1 = 16 - 12 = \underline{4A}$

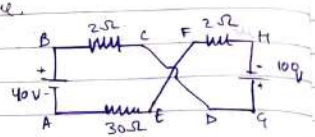
$\therefore R_N = \frac{230}{4} = 57.5\,\Omega$ // Ans.

Q. A wire of $100\,\Omega$ resistance is cut into how many peices so that when they are connected is parallel the resultant (R_{eq}) is $1\,\Omega$.

Ans. $R_{eq} = \frac{100}{n} = n = 10$

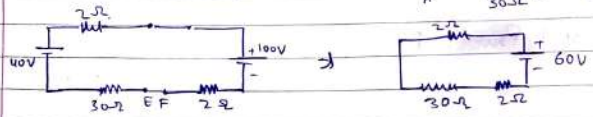
$\frac{100}{n} = \frac{100}{\frac{10}{10}} = 10$

Q. Find the Current I and Voltage V across $30\,\Omega$ resistor of the circuit Showing the figure.



Voltage = 100 − 40

= 60 V



$I = \frac{60}{36} = 1.764\,A$ //

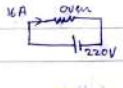
$V = 36 \times 1.764 = 52.92\,V$ //

Q. An oven takes 16A at 220V it is desired to reduce the Current at 12A. Find

(i) The resistance that must be connected in series.

(ii) Voltage across resistance

Ans. Given: Initial Current = 16A, Voltage = 220V., desired Current = 12A



Acc. by Ohm's Law,

$R_1 = \frac{V}{I_1} = \frac{220}{16} = 13.75\,\Omega$

$R_2 = \frac{V}{I_2} = \frac{220}{12} = 18.33\,\Omega$

(i) The Resistance that must be connected in series,

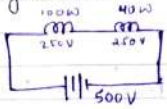
$R = R_2 - R_1 \Rightarrow 18.33 - 13.75$

$R = 4.58\,\Omega$ //

(ii) Voltage across resistance: $V = I_2 \times R$

$V = 12 \times 4.58 = 54.96\,V$ //

Q. A 100 watt 250 Volt bulb is put in series with a 40 watt 250 Volt bulb across 500 V supply. What will be the Current drawn, what will be the power consumed by the bulb? will Such a combination works.



Ans. Given: Power 1: 100 Watt, $V_1 = 250\,V$ — (1)

Power 2: 40 watt, $V_2 = 250\,V$ — (2)

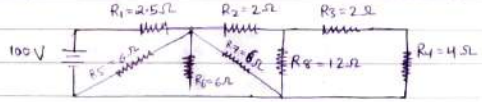
$V_s = 500\,V$

$I_1 = \frac{Power_1}{V_1} = \frac{100}{250} = \frac{2}{5} = 0.4\,A$

$I_2 = \frac{Power_2}{V_2} = \frac{40}{250} = 0.16\,A$

Although they are connected in series current flowing is different which is not possible.

Q. Determine the Current in the circuit shown in the figure.



Ans. R_9:

R_3 & R_4 in Series: $2 + 4 = 6\,\Omega$

R_{10}: R_9 & R_8 in parallel: $\frac{1}{6} + \frac{1}{12} = \frac{6+12}{6\times12} = \frac{18}{72}$ $\frac{1}{4} = 4\,\Omega$

R_{11}: R_{10} & R_2 in Series: $4 + 2 = 6\,\Omega$

R_{12}: R_{11} & R_7 in parallel: $R = \frac{6\times6}{6+6} = \frac{36}{12} = 3\,\Omega$

R_{13}: R_{12} & R_6 in parallel: $R = \frac{6\times3}{6+3} = \frac{18}{9} = 2\,\Omega$

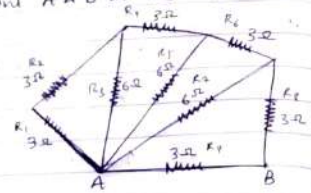
R_{iv} : R_{13} & R_4 in parallel : $R = \frac{2\times6}{2+6} = \frac{12}{8} = \frac{3}{2} = \boxed{1.5\Omega}$

R_i & R_{iv} are in series,

$\therefore R_{eq} = R_i + R_{iv} = 2.5 + 1.5 = \boxed{4\Omega}$

I = total current $= \frac{V}{R} = \frac{100}{4} = 25A$

Q. Calculate the resistance in the point A & B in the network shown in the figure.



Ans → R_a : R_2 & R_3 in series : $3+3 = \underline{6\Omega}$

R_b : R_a & R_3 in parallel : $\frac{6\times6}{6+6} = \underline{3\Omega}$

R_c : R_b & R_4 in series : $3+3 = \underline{6\Omega}$

R_d : R_c & R_5 in parallel : $\frac{6\times6}{6+6} = \underline{3\Omega}$

R_e : R_d & R_6 in series : $3+3 = 6\Omega$

R_f : R_e & R_7 in parallel $= \frac{6\times6}{6+6} = \underline{3\Omega}$

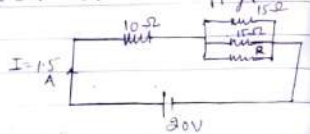
R_g : R_f & R_8 in series : $3+3 = \underline{6\Omega}$

R_h : R_g & R_1 in parallel : $\frac{3\times6}{3+6} = \frac{18}{9} = \underline{2\Omega}$

Q. A resistance of 10Ω is connect in series with two resistance each of 15Ω arranged in parallel. What resistance must be connected across with parallel combination ~~tat~~ that a total current taken shall be 1.5A with 30V supply.

Ans. $R_{eq} = 10 + [15 \| 15 \| R]$

$= 10 + [7.5 \| R]$



$I = 1.5A$

$V = 30V$.

$R_{eq} = \frac{V}{I} = 10 + \left[\frac{7.5\times R}{7.5+R}\right] = \frac{30}{1.5} = \frac{20\times2}{3} = \frac{40}{3}$

$\Rightarrow \frac{7.5R}{7.5+R} = \frac{40}{3} - 10 \Rightarrow \frac{7.5R}{7.5+R} = \frac{10}{3}$

$\Rightarrow 3(7.5R) = 10(7.5+R)$

$22.5R = 75 + 10R$

$12.5R = 75$

$R = \frac{750}{125} = \underline{6\Omega}$ Ans

28/8/24

Q. An aluminium, ~~and~~ ~~+~~ or a copper wire are connected ~~together~~ in parallel. The current flowing in the respective wire is in the ratio of 30:40 the aluminium wire is 45% longer than the copper wire and the ratio of their specific resistance is 30:17. Find the ratio of their cross-sectional area.

Ans → $Al = 45\% + Cu \Rightarrow l_{Al} = 0.45 l_{Cu} + l_{Cu} = \underline{1.45 l_{Cu}}$

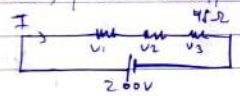
$\frac{I_{Al}}{I_{Cu}} = \frac{30}{40} \Rightarrow \frac{R_{Al}}{R_{Cu}} = \frac{40}{30}$, $\frac{\rho_{Al}}{\rho_{Cu}} = \frac{30}{17}$

$R = \frac{\rho l}{a} \Rightarrow \frac{R_{Al}}{R_{Cu}} = \frac{\rho_{Al}}{\rho_{Cu}} \times \frac{1.45 l_{Cu}/A_{Al}}{l_{Cu}/A_{Cu}}$

$\Rightarrow \frac{40}{30} = \frac{30}{17} \times \frac{1.45 \times A_{Cu}}{A_{Al}}$

$\Rightarrow \frac{A_{Al}}{A_{Cu}} = \frac{30}{17} \times \frac{3}{4} \times 1.45 = \underline{1.91 : 1}$

Q. 3 resistances 30Ω, 25Ω, 45Ω are connected in series across 200V. Calculate Total resistance, current, potential diff. across each element.



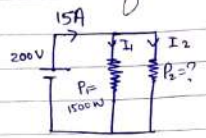
Ans → (i) $R_{eq} = R_1 + R_2 + R_3 = \underline{100\Omega}$

(ii) Current : $I = \frac{V}{R} = \frac{200}{100} = \underline{2A}$

(iii) Potential diff. :- $V_1 = I\times30 = 2\times30 = \underline{60V}$, $V_2 = I\times25 = 2\times25 = \underline{50V}$, $V_3 = I\times45 = 2\times45 = \underline{90V}$

Q. 2 Resistances are connected in parallel and ~~that~~ a Voltage of 200V is applied to the terminal, The total current is 15 A and the power dissipated in one of the resistor is 1500 watt. What is the resistance of each resistor.

Ans



$P_1 = \frac{V^2}{R_1} = 1500 = \frac{1200^2}{R_1}$

$= R_1 = \frac{40000}{1500} = 26.67\ \Omega$

$I_1 = \frac{V}{R_1} = \frac{200}{26.07} = 7.5\ A$

$I = I_1 + I_2 \rightarrow I_2 = I - I_1$

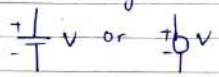
$= I_2 = 15 - 7.5 = 7.5\ A$

then, $R_2 = \frac{V}{I_2} = \frac{200}{7.5} = 26.67\ \Omega$ //

Sources : 1) Independent Sources 2) dependent Sources

- Independent Sources : The element for which both Voltage and current do not depend Voltage or current ~~as the other~~ elsewhere in the Circuit.

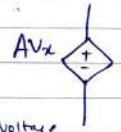
Voltage Source



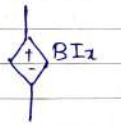
Current Source.

- dependent Sources : The element for which either the Voltage or current depends upon the Voltage or current elsewhere in the Circuit.

Voltage Source

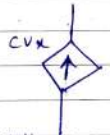


Voltage ~~Current~~ controlled Voltage Source

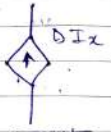


current controlled Voltage Source

Current Source

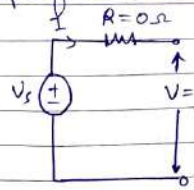


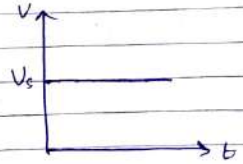
Voltage Controlled Current Source



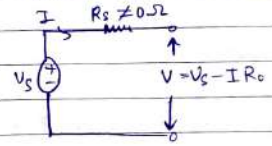
current controlled current Source.

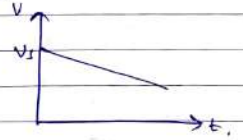
- Ideal Voltage Source : It is a two terminal element with the property that the Voltage across the terminal is specified and every instant in time.



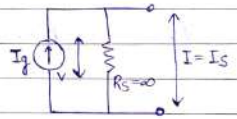


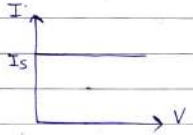
- Practical Voltage Source : A Voltage Source which has a infinite internal resistance and whose voltage drops with increase in current drawn from it is known as Practical or Real Voltage Source.



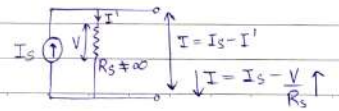


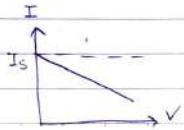
- Ideal Current Source : It is a circuit element that maintains a prescribed current through its terminals regardless of the voltage across these terminals is called ideal current source.





- Practical current Source : A practical current source is equivalent to an ideal current in parallel with high resistance or low conductance.





Node, Loop, and Branch

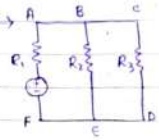
↳ **Node** :- When two or more branches are connected together then the common point is called as node

They are of two types :-

1) Simple Node – When two branches are connected together then the common point is known as simple node (no current division)

2) Principle Node – When more than two branches are connected together then the common point is known as principle node (current division takes place).

Example,



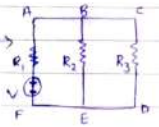
Node → A, B, C, D, E, F

↳ Simple node :- A, C, D, F

↳ Principle node :- B and E

↳ **Loop** :- The starting node and ending node of a path is same.

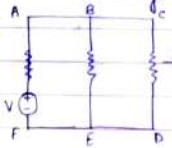
Example,



Loop → ABEFA, BCDEB, ACDFA

↳ **Branch** :- If the several element in a network carry a same current then they can referred as a branch.

Example,



Branches → AB, BE, CD, ED, EFAB, BCDE

(ABC is not a branch)

Kirchoff's Law :-

The amount of current flowing into a node is equal to the sum of currents flowing out of it.

⇒ KCL (Kirchoff's current law) :- The algebraic sum of all the currents at a given node is zero

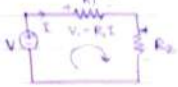
$$\boxed{\Sigma I = 0}$$

$I_1 - I_2 + I_3 + I_4 - I_5 = 0$ or $I_1 + I_3 + I_4 = I_2 + I_5$

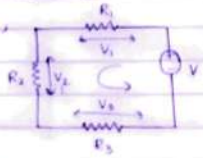
⇒ KVL (Kirchoff's voltage law) :- The algebraic sum of all the voltage in the closed loop is zero.

$$\Sigma V = 0$$

$$\boxed{V - IR_1 - IR_2 = 0}$$



For Example,

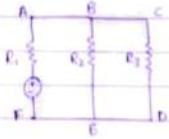


$$\boxed{V - IR_1 - IR_2 - IR_3 = 0}$$

Mesh Analysis :-

It is a loop which does not contain any inner loop.

Example,



l_1 = ABEFA
l_2 = BCDEB } Mesh

l_3 = ACDFA } does not count as a mesh

Procedure :-

↳ Step 1 :- Identify the total no of meshes

Step 2 :- Assign mesh current.

Step 3 :- Develop the KVL equation for each mesh

Step 4 :- Solve the KVL equation to find the mesh currents

Q Find the power loss in 10 Ω resistor

Ans

$R_{eq} = 5 + 5 \| 10 = 5 + \frac{50}{15} = \boxed{8.33\ \Omega}$

$V = I R_{eq}$ $\therefore I = \frac{V}{R_{eq}} = \frac{10}{8.33} = \boxed{1.2\,A}$

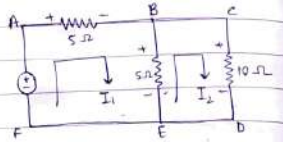
Now $I = I_1 + I_2$ (current divider rule)

$\therefore I_2 = \frac{I \cdot 5}{5+10} = \frac{1.2 \times 5}{15} = \boxed{0.4\,A}$

$I_1 = \frac{I \cdot 10}{5+10} = \frac{1.2 \times 10}{15} = \boxed{0.8\,A}$

$\therefore I = I_1 + I_2 = 0.8 + 0.4 = \boxed{1.2\,A}$ Hence verified

Mesh Analysis :-



Step 1) No of mesh = 2

Step 2) Current Element = I_1, I_2

Step 3) Equations :-

$10 - 5I_1 - 5(I_1 - I_2) = 0$

$10 - 5I_1 - 5I_1 + 5I_2 = 0$

$10 - 10I_1 + 5I_2$ —— ①

and $-10I_2 - 5(I_2 - I_1) = 0$

$-10I_2 - 5I_2 + 5I_1 = 0$

$-15I_2 + 5I_1 = 0$ —— ②

Step 4) Solving the equation

$-10I_1 + 5I_2 = -10$

$(5I_1 - 15I_2 = 0) \times 2$

$-10I_1 + 5I_2 = -10$

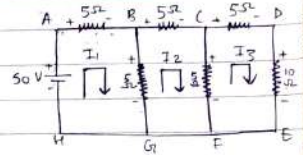
$10I_1 - 30I_2 = 0$

$-25I_2 = -10$

$I_2 = \frac{10}{25} = \boxed{0.4\,A}$ $\therefore I_1 = \boxed{1.2\,A}$ Total current

Now, Power in 10Ω resistor $= I^2R = (0.4)^2 \times 10 = 0.16 \times 10$
$= \boxed{1.6\text{ Watt}}$

Q. Find the voltage drop in 10Ω resistor?



Step 2: No. of meshes = 3

Step 2: Current element = I_1, I_2, I_3

Step 3: equations :

Loop 1 : $50 - 5I_1 - 5(I_1 - I_2) = 0$

$50 - 10I_1 + 5I_2 = 0$ —— ①

Loop 2 : $-5I_2 - 5(I_2 - I_3) - 5(I_2 - I_1) = 0$

$+5I_1 - 15I_2 + 5I_3 = 0$ —— ②

Loop 3 : $-5I_3 - 10I_3 - 5(I_3 - I_2) = 0$

$5I_2 - 20I_3 = 0$

$I_2 - 4I_3 = 0$ —— ③

$I_2 = 4I_3$ —— ④

Substituting in eq ②,

$5I_1 - 60I_3 + 5I_3 = 0$

$5I_1 - 55I_3 = 0$

$I_1 - 11I_3 = 0$

$I_1 = 11I_3$ —— ⑤

Substituting in eq ①,

$50 - 10I_1 + 5I_2 = 0$

$50 - 10(11I_3) + 20I_3 = 0$

$50 - 110I_3 + 20I_3 = 0$

$I_3 = 5/9\ A$

Putting the value of I_3 in eq. I_1 & I_2,

$I_1 = 11 \times \frac{5}{9} = \frac{55}{9}\ A$, $I_2 = 4 \times \frac{5}{9} = \frac{20}{9}\ A$

Potential drop in 10Ω resistor : $V_{10} = IR$

$= I_3 \times 10 = \frac{5}{9} \times 10 = \frac{50}{9}\ V$ ~5.5V

By ohm's law:

$5 + 5 \| [5 + 15 \| 5] \rightarrow \frac{20+15}{4} = \frac{35}{4}$

$5 + [5 \| 8.75] \rightarrow \frac{5 \times 8.75}{13.75}$

$5 +$

$R_{eq} = 8.18 \Rightarrow V = IR \Rightarrow 50 = I \times 8.18$

$\Rightarrow I = 6.112$

$I_a = \frac{6.13 \times 5}{13.84}$, $I_a = 2.21A$, $I_1 = \frac{55}{9} = 2.21A$.

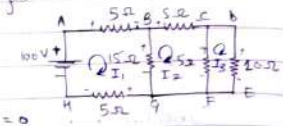
$I_c = \frac{2.21 \times 5}{20} = 0.55A$, $I_3 = \frac{5}{9} = 0.55A$.

$V = I_c \times 10$

$0.55 \times 10 = 5.5V$

f. find the value of current through 15Ω resistance.

Ans → no of mesh: 3



Current element I_1, I_2, I_3

equation

loop 1: $100 - 5I_1 - 15(I_1 - I_2) - 5I_1 = 0$

$100 - 5I_1 - 15I_1 + 15I_2 - 5I_1 = 0$

$100 - 25I_1 + 15I_2 = 0$ — (1)

loop 2: $-5I_2 - 5(I_2 - I_3) - 15(I_2 - I_1) = 0$

$-5I_2 - 5I_2 + 5I_3 - 15I_2 + 15I_1 = 0$

$-25I_2 + 5I_3 + 15I_1 = 0$ — (2)

loop 3: $-10I_3 - 5(I_3 - I_2) = 0$

$-10I_3 - 5I_3 + 5I_2 = 0$

$-15I_3 + 5I_2 = 0$ — (3)

$I_2 = \frac{15}{5} I_3 = \boxed{I_2 = 3I_3}$ — (4)

Substituting eq 4 in 2

$-25(3I_3) + 5I_3 + 15I_1 = 0$

$-75I_3 + 5I_3 + 15I_1 = 0$

$-70I_3 + 15I_1 = 0 \Rightarrow I_1 = \frac{70}{15} I_3 \sim 4.66$ — (5)

Substituting eq 4, 5 in 1

$100 - 25(70/15\, I_3) + 15(3I_3) = 0$

$100 - \frac{1750}{15} I_3 + \frac{45}{1} I_3 = 0 \Rightarrow 100 - \left(\frac{1750 + 675}{15}\right) I_3 = 0$

$\Rightarrow 100 - \frac{1075}{15} I_3 = 0 \Rightarrow * I_3 = \frac{100 \times 15}{1075} = \frac{1500}{1075} \sim 1.39A$

Putting the value of I_3 in eq 4, 5 :

* $I_1 = \frac{70}{15} I_3 \Rightarrow \frac{70}{15} \times \frac{1500}{1075} = 4.66 \times 1.39$
$= 6.51A$

* $I_2 = 3I_3 = 3 \times \frac{1500}{1075} = 3 \times 1.39$
$= 4.17A$

∴ Current passing through 15Ω resistor $= I_1 - I_2$
$= 6.44 - 4.17 = 2.3Ω$

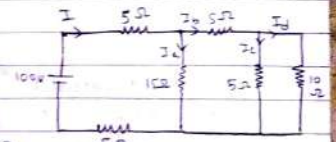
by Ohms law,

$R_{eq} = 5 + 5 + [15 \| [5 + 5 \| 10]]$

$5 + 5 + [15 \| [5 + \frac{10}{3}]]$

$5 + 5 + [15 \| (15/3)] = 8.33$

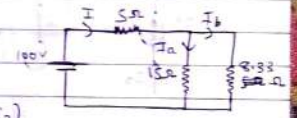
$5 + 5 + 5.3 = 15.35Ω$



$I = \frac{V}{R} = \frac{100}{15.35} = 6.51$

Current divider rule,

$I = I_a + I_b$, $I_b = I_c + I_d$



$I_a = \frac{6.51 \times 8.33}{23.33} = 2.32A \ (= I_1 - I_2)$

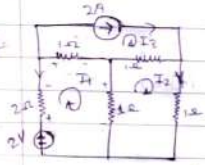
15+8.33
23.33

$I_b = \frac{6.51 \times 15}{23.33} = 4.18A$, $I_c = \frac{4.18 \times 10}{15} = 2.78A$

$I_d = \frac{4.18 \times 5}{15} = 1.3A \ (= I_2 - I_3)$

4/9/24

f.



No. of mesh = 7

Current element = I_1, I_2, I_3

equations: loop 1: $2 - 2I_1 - 1(I_1 - 2I_3) - 1(I_1 - I_2) = 0$

$4 - 4I_1 + I_2 = 0$ — (1)

loop 2: $-(I_2 - 2) - I_2 - 1(I_2 - I_1) = 0$

$I_1 - 3I_2 + 2 = 0$ — (2)

Multiply eq (1) by 3,

$4 - 4I_1 + I_2 = 0 \times 3 \Rightarrow -12I_1 + 3I_2 + 12 = 0$ — (3)

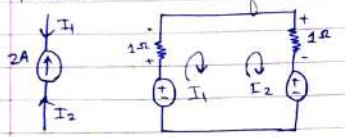
div. eq. (3) & (2),

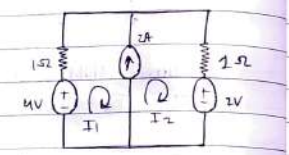
$$-12I_1 + 3I_2 = -12$$
$$I_1 - 3I_2 = -2$$
$$-11I_1 = -14$$

$\therefore I_1 = \frac{14}{11}, \quad I_2 = \frac{12}{11}$

- Super Mesh Analysis:
When a current source is present between two meshes we remove the branch having a current source and then the remaining loop is known as super mesh.

Q. Find out the value of current.





$I_2 - I_1 = 2$ — (2)

$4 - I_1 - I_2 - 2 = 0$

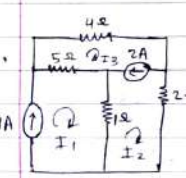
$2 - I_1 - I_2 = 0$ — (1)

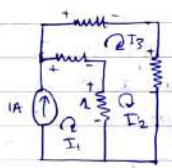
$I_2 = 2 + I_1 \Rightarrow 2 - I_1 - (2 + I_1)$

$= 2 - I_1 - 2 - I_1 = 2I_1 = 0 = \boxed{I_1 = 0}$

$\boxed{I_2 = 2}$

Q.

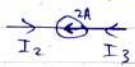




$-5(1 - I_3) - 1(1 - I_2) = 0$ — (1)

$-5(1 - I_3) - 1 + I_2 = 0$

$-4(I_3) - 2(I_2 - 1)(I_2 - 1) - 5(I_3 - 1) = 0$ — (2)



$I_3 - I_2 = 2$

$I_2 = I_3 - 2$ — (3)

eq-(2) : $-4(I_3) - 2I_2 - 1(I_2 - 1) - 5(I_3 - 1) = 0$

$-4I_3 - 2I_2 - I_2 + 1 - 5I_3 + 5 = 0$

$-4I_3 - 3I_2 + 6 - 5I_3 = 0$

$-3I_2 - 9I_3 + 6 = 0$ — (2)

Putting eq 3 in 2. : $-3(I_3 - 2) - 9I_3 + 6 = 0$

$-3I_3 + 6 - 9I_3 + 6 = 0$

$-12I_3 + 12 = 0$

$I_3 = \frac{12}{12} = \boxed{1}$

$I_2 = I_3 - 2$

$I_2 = 1 - 2 = \boxed{-1}$

Putting eq 3 in 1 : $-5 + 5I_3 - 1 + I_2 = 0$

$-6 + 5I_3 + I_2 = 0$

$I_2 = -5I_3 + 6$

$I_3 - 2 + 5I_3 - 6 = 0$

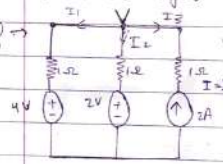
$-8 + 6I_3 = 0$

$I_3 = \frac{8}{6} = \boxed{\frac{4}{3}}$

5/9/20

(*) Nodal Analysis:

Q.

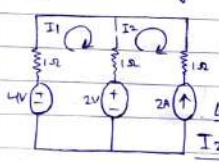


$V = I_1 + I_2 + I_3 = 0$

$I = \frac{V}{R} \Rightarrow \frac{V-4}{1} + \frac{V-2}{1} - 2 = 0 \Rightarrow V = 4V$

$I_1 = \frac{V-4}{1} = 0A, \quad I_2 = \frac{V-2}{1} = 2A, \quad I_3 = -2A$

By mesh analysis:



Loop1: $+4 - I_1 - I_1 + I_2 - 2 = 0$

$2I_1 + I_2 + 2 = 0$

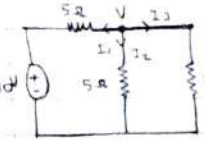
$2I_1 + (-2) + 2 = 0$

$2I_1 = 0$

$\boxed{I_1 = 0A}$

Loop2: $I_2 = -2A$

Q.

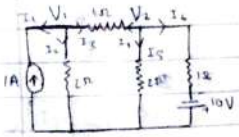


$V = I_1 + I_2 + I_3 = 0$

$I = \frac{V}{R} \Rightarrow \frac{V-10}{5} + \frac{V}{5} + \frac{V}{10} = 0 \Rightarrow V =$

$I_1 = -1.2A$, $I_2 = 0.8A$, $I_3 = 0.4A$

Q



$V_1 = I_1 + I_2 + I_3 = 0$

$-1 + \frac{V_1}{2} + \frac{V_1 - V_2}{1} = 0$

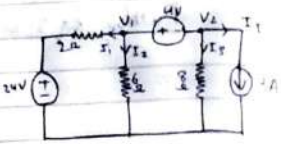
$V_2 = I_4 + I_5 + I_6 = 0$

$\frac{V_2 - V_1}{1} + \frac{V_2}{2} + \frac{V_2 + 10}{1} = 0$

$I_3 = -I_1$

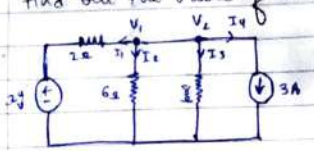
$I_1 = -1$

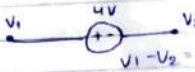
6/9/24 Super Node -



Q find out the value of current -

Ans.





$V_1 - V_2 = 4$

$V_1 = 4 + V_2$

According to KCL,

$I_1 + I_2 + I_3 + I_4 = 0$

$\frac{V_1 - 24}{2} + \frac{V_1}{6} + \frac{V_2}{8} + 3 = 0$

$\frac{12V_1 - 288 + 4V_1 + 3V_2 + 72}{24} = 0$

$16V_1 + 3V_2 - 216 = 0$ — ①

$16(4 + V_2) + 3V_2 - 216 = 0$

$64 + 16V_2 + 3V_2 - 216 = 0$

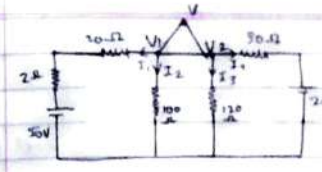
$19V_2 = 152$

$\boxed{V_2 = 8V}$

$V_1 = 4 + V_2 = 4 + 8 = \boxed{12V}$

$I_1 = \frac{12 - 24}{2} = -\frac{12}{2} = \boxed{-6A}$ (opposite direction), $I_2 = \frac{12}{6} = \boxed{2A}$, $I_3 = \frac{8}{8} = \boxed{1A}$

Q.



$I_1 + I_2 + I_3 + I_4 = 0$

$\frac{V_1 - 50}{2} + \frac{V}{100} + \frac{V}{120} + \frac{V - 20}{20} = 0$

$\Rightarrow \frac{300V - 15000 + 66V + 55V + 220V - 4400}{6600} = 0$

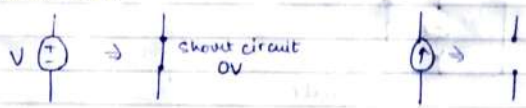
$\Rightarrow \frac{641V - 19400}{6600} = 0 \Rightarrow V = \frac{19400}{641} = 30.26V$

$I_1 = \frac{30.26 - 50}{2} = \frac{-19.74}{2} = \boxed{-0.99}$, $I_2 = \frac{30.26}{100} = \boxed{0.3026A}$

$I_3 = \frac{30.26}{120} = \boxed{0.25A}$, $I_4 = \frac{30.26 - 20}{20} = \frac{10.26}{20} = \boxed{0.512}$

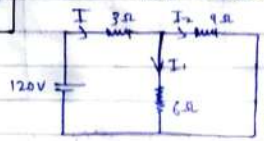
11/9/24

→ Superposition Theorem: If there are two or more than two voltage source acting simultaneously the current flowing through any section is the algebric sum of the current which should flow in that section if each source were considered separately and all the other sources are replaced for the time being by their internal resistance. It means all the dependent sources are replaced by the internal resistances i.e. we replace every voltage source by 0V and every current source 0A.



Q. Find the value of current through 6Ω resistance by superposition theo

Considering 120V as a voltage and 60V is short circuited.



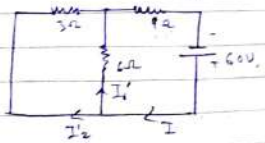
$I = V/R_{eq} = \frac{120}{\cancel{8.7}}$ ~~10A~~

$R_{eq} = 3 + 6 \| 9 =$

By current divider rule,

$I_1 = \frac{18.18 \times 9}{6+9} = 10.9A$

Now, considering 60V source and 120 V is short circuited



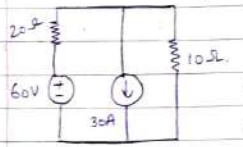
$R_{eq} = 3 \| 6 + 9 = 11$

$I = \frac{60}{11} = 5.45A$

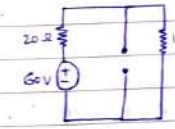
$I_1' = \frac{5.45 \times 3}{9} = 1.81 A$

$I_1 - I_1' = 10.9 - 1.81 = 9.09A$

Q. Find the value of current through 10Ω resistance.



Consider 60V source and 30A is open



$R_{eq} = 20 + 10 = 30 \Omega$

$I = \frac{V}{R} = \frac{60}{30} = 2A$

Now, considering current 30A and 60V is short circuited

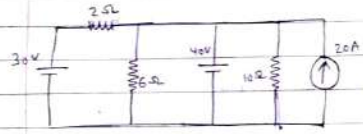
Current divider rule,

$I' = \frac{30 \times 20}{30} = 20A$

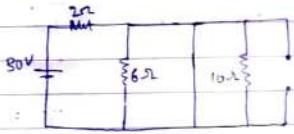
Total current ⇒ $I' - I = 20 - 2 = 18A$

12/7/24

Q. Using superposition thm find the value of current through 10Ω resistance.



Consider 30V source & 40V is short circuited & 20A is open

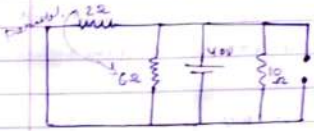


$R_{eq} = 2 + (6 \| 10) = 28/4$

$I = \frac{V}{R} = \frac{30 \times 4}{23} = 5.21A$

Since short circuit is in parallel all current will pass through it. No current will pass from 10Ω resistance.

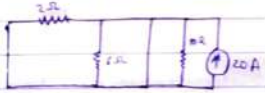
Consider 40V source & 30V short circuited & 20A is open



$R_{eq} = \frac{40}{9} \Omega$ $\left[2 \| 6 + 10 = 10 \right]$

$I = \frac{V}{R} = \frac{40}{10} = 4A$

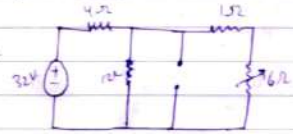
Consider current 20A & 30V & 40V source is short circuited



Short circuit is in parallel. Current across circuit is 0.

Q. Find the value of current through 6Ω

Consider 32V source & 2A is open



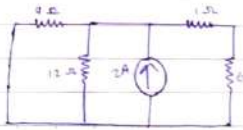
$R_{eq} = 4 + (12 + 1 \| (6+1))$

$= 4 + 12 \| 7 = 160/19$

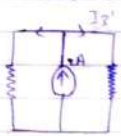
$I = \frac{V}{R} = \frac{32 \times 19}{160} = \frac{19}{5} = 3.8A$

By current divider rule, $\frac{3.8 \times 12}{12+7} = 2.4A$

Consider 2A current and 32 V is short circuited



⇒

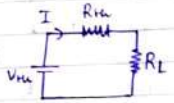


$I_3' = \frac{2 \times 3}{10} = 0.6$

$I = 2.4 + 0.6 = 3A$

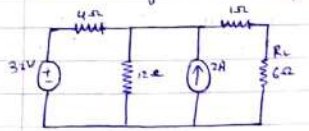
→ Thevenin Theorem: The current flowing through a resistor connected across any two terminals of a network by an equivalent circuit having a voltage source V_{th} in series with a resistor R_{th}. Where, V_{th} is the open circuit voltage between the required two terminals called Thevenin voltage and R_{th} is the equivalent resistance of the network as seen from the two terminal with all other sources replaced by their internal resistance called Thevenin resistance.

Thevenin equivalent circuit:

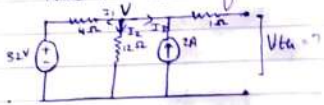


$$I = \frac{V_{th}}{R_{th} + R_L}$$

Q. Find value of current through 6Ω resistance by Thevenin Theorem.



Firstly, remove load branch and find the value of V_{th}



By Nodal Analysis,

$V = I_1 + I_2 + I_3 = 0$

$= \frac{V-32}{4} + \frac{V}{12} - 2 = 0 \Rightarrow 3V - 96 + V - 24 = 0$

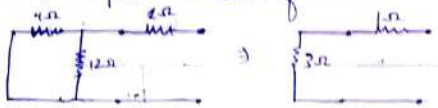
$4V - 120 = 0 \Rightarrow \boxed{V = 30V}$

$I_2 = \frac{V}{12} = \frac{30}{12} = 2.5$

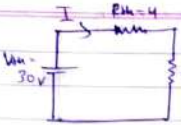
$\Rightarrow V_{12} \Rightarrow I_2 \times 12 = \frac{30 \times 12}{12} = 30V$

By Acc. to Thevenin Thm. the value of V_{th} will be 30V.

Then, To find the value of R_{th}:

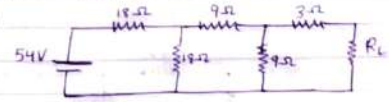


$\Rightarrow R_{th} = 4\Omega$

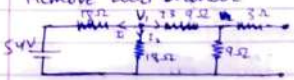


$$I = \frac{V_{th}}{R_{th} + R_L} = \frac{30}{4+6} = 3A.$$

Q. For the given circuit find the value of current through R_L.
$R_L \Rightarrow$ (i) 3Ω, (ii) 6Ω, (iii) 9Ω



(i) Remove load branch



By nodal analysis

$V = I_1 + I_2 + I_3$

$= \frac{V-54}{18} + \frac{V}{18} + \frac{V}{18} = 0 \Rightarrow 3V - 54 = 0$

$\boxed{V = 18V}$

$I_3 = \frac{18}{18} = 1$, $V_{th} = 9 \times 1 = \boxed{9V}$

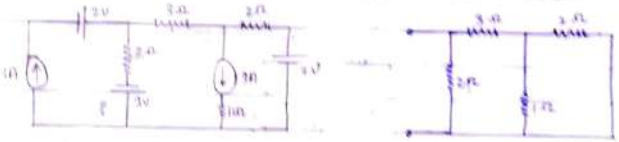
$R_{th} : \rightarrow [[(18 \| 18) + 9) \| 9] + 3$

$= 9\Omega$

$\therefore I = \frac{V_{th}}{R_{th} + R_L} = \frac{9}{9+3} = 0.7$ (i), $\frac{9}{9+6} = 0.6$ (ii), $\frac{9}{9+9} = 0.5$ (iii)

16/9/24

Q. Using Superposition theorem, find the value of current across 3Ω resistance

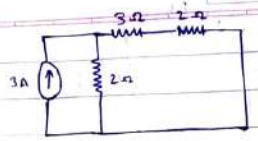


In 2V no current is flowing in the circuit.

Considering 3A,

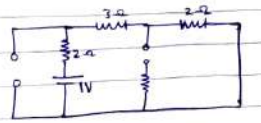
Current divider:

$I = \frac{3 \times 2}{7} = \frac{6}{7}$



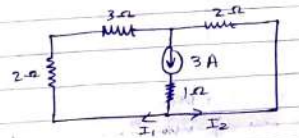
Considering 1V

$I = \frac{1}{7}$



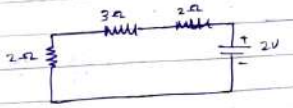
Considering 3A,

$I_1 = \frac{3 \times 2}{7} = \frac{6}{7}$



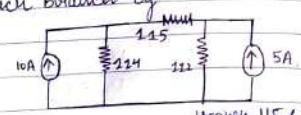
Considering 2V,

$I = \frac{2}{7}$

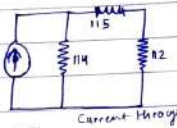


Total Current $= \frac{6}{7} + \frac{6}{7} - \frac{2}{7} - \frac{1}{7} = \frac{9}{7}$ A

Q. find the value of current in each branch by -
- Superposition
- Mesh Analysis
- Nodal Analysis.



Ans → Consider 10A,



Through 115 & 112Ω:

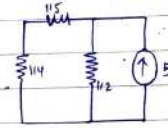
By current divider,

$I = \frac{10 \times 114}{341} = \frac{1140}{341}$

~ 3.634 A

Current through 114 $= I_2 = \frac{10 \times 227}{341} = 2270/341 = 6.65$ A

Consider 5A,



$I' = \frac{5 \times 112}{341} = \frac{560}{341}$

~ 1.64

Current through 115 & 114Ω

Total current through 112Ω resistor: $I = \frac{5 \times 229}{341} = \frac{114}{5} = 3.35$ A

↳ Total Current through 112 → $3.35 + 3.34 = 6.69$ A

↳ Total Current through 114 → $6.65 + 1.64 = 8.29$ A

↳ Total Current through 115 → $3.34 + 1.64 = 1.70$ A

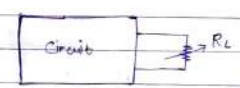
27/9/24

• NORTAN'S THEOREM:

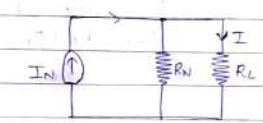
A linear two terminal network can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistance R_N.

Where In is the short circuit current supplied by the source that would flow between the two selected terminals when they are short circuited. It is generally called Nortan's Current.

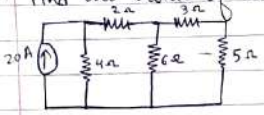
R_N is the equivalent resistance of the network as seen from the two terminals with all the other voltage source replaced by short circuit and current sources replaced by open circuit. It is generally called Nortan's resistance.



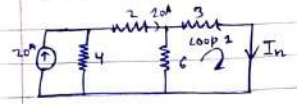
$I = \frac{I_N \times R_N}{R_N + R_L}$



Q.. Find the value of current through 5Ω resistor by Norton.

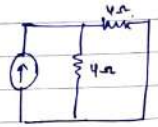


First short circuit 5Ω resistance.



Loop 1 = $\frac{1}{3} + \frac{1}{6} = \frac{1}{R_{eq}}$.

$\frac{2+1}{6} \Rightarrow$ ~~Req~~ = 2Ω.



Loop 2 -

2+2 = 4Ω — Req.

By Current divider $\Rightarrow I_1 = \frac{20 \times 4}{8} = 10\ A$

again, $\Rightarrow I_n = \frac{10 \times 6^2}{9_3} = \frac{20}{3}\ A \sim 6.66A$

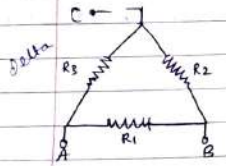
$R_n = [2+4 \| 6 (+3)]$

$[6 \| 6] + 3$

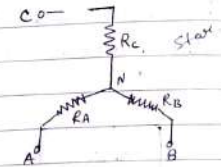
$3 + 3 = 6$

19/9/24

· Star-delta Transformation



Delta to Star



disconnect C,

R_3 & R_2 in Series, parallel with R_1.

$R_{AB} = R_1 \| (R_3 + R_2)$

$= \frac{R_1R_2 + R_1R_3}{R_1+R_2+R_3}$ — (1)

disconnect C,

R_C is dummy & R_A, R_B in Series

$R_{AB} = R_A + R_B$

$= R_A + R_B$ — (1)

disconnecting A,

$R_{BC} = R_2 \| (R_1 + R_3) = R_B + R_C$ — (2)

$= \frac{R_1R_2 + R_2R_3}{R_1+R_2+R_3}$ — (2)

disconnecting B,

$R_{AC} = R_3 \| (R_1 + R_2) = R_A + R_C$ — (3)

$= \frac{R_1R_3 + R_2R_3}{R_1+R_2+R_3}$

By adding eq - 1, 2, 3.

$\cancel{2}(R_A + R_B + R_C) = \frac{\cancel{2}(R_1R_2 + R_2R_3 + R_3R_1)}{R_1+R_2+R_3}$.

$\Rightarrow R_A + R_B + R_C = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1+R_2+R_3}$ — (4)

Subtract: eq 4 ~~from~~ -1,

$R_A + R_B + R_C = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1+R_2+R_3} - \frac{R_1R_2 + R_1R_3}{R_1+R_2+R_3} = R_A + R_3$

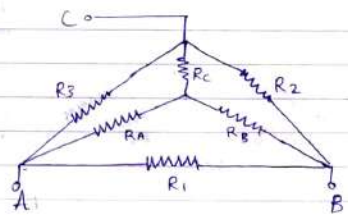
$\Rightarrow R_C = \frac{R_2R_3}{R_1+R_2+R_3}$ — (5)

eq 4 - 2, $R_A = \frac{R_1R_3}{R_1+R_2+R_3}$ — (6)

eq - 4 - 3, $R_B = \frac{R_1R_2}{R_1+R_2+R_3}$ — (7)

$R_C = \frac{R^2}{3R} = \frac{R}{3}$

$R_{Star} = \frac{R_\Delta}{3}$



(multiply opp. branches)
(and divide by $R_1+R_2+R_3$)

Star to delta: eq: 5 × 6,

$$R_A R_C = \frac{R_2 R_3 \; R_1 R_3}{(R_1+R_2+R_3)^2} \quad \text{— (8)}$$

eq 6 × 7,

$$R_A R_B = \frac{R_1 R_3 \; R_1 R_2}{(R_1+R_2+R_3)^2} \quad \text{— (9)}$$

eq. 7 × 5,

$$R_B R_C = \frac{R_1 R_2 \cdot R_2 R_3}{(R_1+R_2+R_3)^2} \quad \text{— (10)}$$

By adding eq 8, 9, 10 we get,

$$R_A R_B + R_B R_C + R_C R_A = \frac{R_1 R_2 R_3 \; \cancel{(R_1+R_2+R_3)}}{(R_1+R_2+R_3)^{\cancel{2}}} \quad \text{— (11)}$$

divide eq 11 ÷ 5,

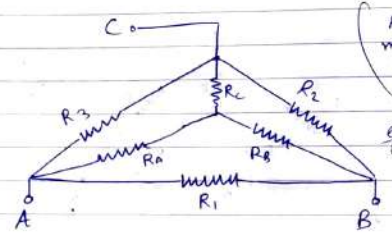
$$R_1 = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C} \rightarrow R_1 = \frac{R_A R_B + R_B + R_A}{R_1}$$

eq 11 ÷ 6,

$$R_2 = \frac{R_B R_C}{R_A} + R_B + R_C$$

eq 11 ÷ 7,

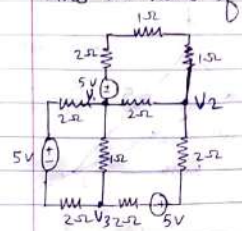
$$R_3 = \frac{R_A R_C}{R_B} + R_A + R_C$$



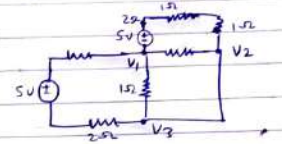
Add opp. branch & multiply opp. branch then div. by 3 branch resistor

eg for R_3: add $R_A + R_C$ multiply $R_A R_C$ & div by R_B

Q Find the value of current through each branch by nodal.



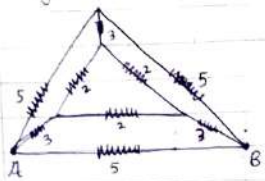
Super node.



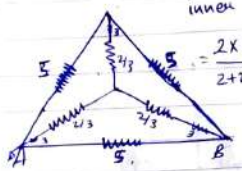
- $\frac{V_1-5}{4} + \frac{V_1+5}{4} + \frac{V_1-V_3}{1} + \frac{V_1-V_2}{2} = 0$
- $\frac{V_2-5}{4} + \frac{V_2-V_1}{2} = 0$
- $V_2 - V_3 - 5 = 0$

(1) → $V_1 - 5 + V_1 + 5 + V_1 - V_3 +$

Q. using Star-Delta find the resistance across A & B.



⇒

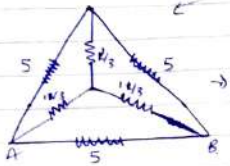


inner delta,

$= \frac{2\times2}{2+2+2} = \frac{4}{6} = \frac{2}{3}$

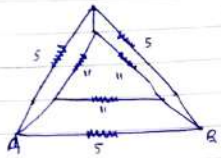
$\frac{2}{3}$ & 8 in series $= \frac{2}{3} + 7 = \frac{11}{3}$

→



→ $\frac{11}{3} + \frac{11}{3} + \frac{\frac{11}{3}\times\frac{11}{3}}{\frac{11}{3}} = 11$

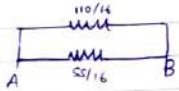
→



5 & 11 are in parallel.

$\frac{5\times11}{5+11} = \frac{55}{16}$

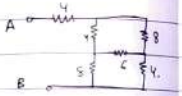
→

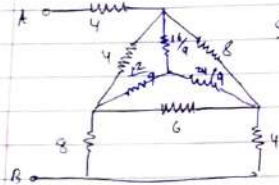


$\frac{110}{16}$ & $\frac{55}{16}$ are in parallel.

$\frac{\frac{110}{16}\times\frac{55}{16}}{\frac{110}{16}+\frac{55}{16}} =$

Q. find the equivalent resistance between A & B.





Solving delta (4, 8, 6)

$\frac{4\times8}{4+8+6} = \frac{32}{18} = \frac{16}{9}$, $\frac{4\times6}{4+6+8} = \frac{24}{18} = \frac{12}{9}$

$\frac{6\times8}{4+6+8} = \frac{48}{18} = \frac{24}{9}$

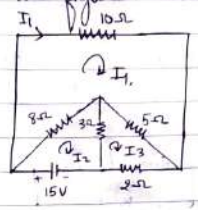
Now, $\frac{12}{9}$ in series with 8 $= \frac{84}{9}$

$\frac{24}{9}$ in series with 4 $= \frac{60}{9}$

parallel $= \frac{39}{5}$

$\frac{39}{5}$ in series with $\frac{52}{9} = \frac{87}{9}$

Q. What is the power loss in 10 Ω resistor in the network shown in the figure. Solve using mesh analysis.



- No. of mesh : 3
- Current element : I_1, I_2, I_3
- equations:

~~1) $15 - 10I_1 - 5I_3 = 0$~~

1) $-10I_1 - 5(I_1 - I_3) - 8(I_1 - I_2) = 0$

$= -23I_1 + 8I_2 + 5I_3 = 0$ — (1)

2) $15 - 8(I_2 - I_1) - 3(I_2 - I_3) = 0$

$= 15 + 8I_1 - 11I_2 + 3I_3 = 0$ — (2)

3) $-5(I_3 - I_1) - 2(I_3) - 3(I_3 - I_2) = 0$

$= 5I_1 + 3I_2 - 10I_3 = 0$ — (3)

On solving eq. 1, 2, 3.

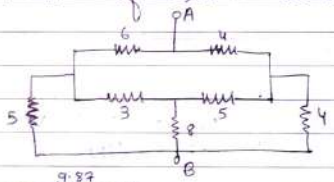
$\boxed{I_1 = 1.23A}$, $I_2 = 41/19\, I_1$, $I_3 = 109/95\, I_1$.

Power $= I^2R$

$= (1.23)^2 \times 10$

$= 15.13$ watt

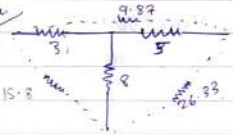
Q. find the value of resistance across A & B.



$3 + 5 + \frac{3\times5}{8}$

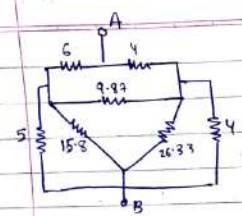
$= 9.87$

Star



$= 3 + 8 + \frac{3\times8}{5}$, $8 + 5 + \frac{8\times5}{3}$

$= 15.8$ $= 26.33$

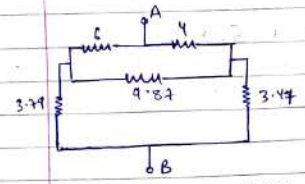


→ 5 & 15.8 in parallel.

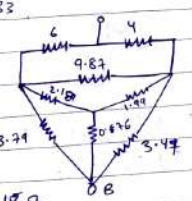
$$\frac{5 \times 15.8}{5+15.8} = \underline{3.79\,\Omega}$$

→ Similarly, 4 & 26.33 in parallel.

$$\frac{4 \times 26.33}{4+26.33} = \underline{3.47\,\Omega}$$



⇒

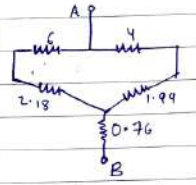


→ $\frac{9.87 \times 3.79}{9.87+3.79+3.41} = \underline{2.18\,\Omega}$

→ $\frac{9.87 \times 3.47}{9.87+3.79+3.41} = \underline{1.99\,\Omega}$

→ $\frac{3.79 \times 3.47}{3.79+3.47+9.87} = \underline{0.876\,\Omega}$

→



→ 2.18 & 6 in series,
$2.18+6 = \underline{8.18\,\Omega}$.

→ 4 & 1.99 in series,
$1.99+4 = \underline{5.99\,\Omega}$.

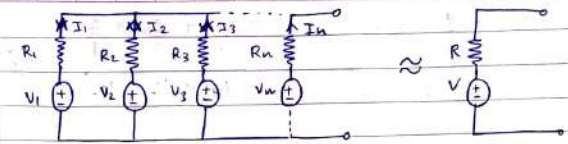
→ 8.18 is in parallel with 5.99 = $\frac{8.18 \times 5.99}{5.99+8.18} = \underline{3.46\,\Omega}$

→ 3.46 & 0.76 in series.
$3.46 + 0.76 = \underline{4.227\,\Omega}$.

25/9/24

* Millmann's Theorem :

(i) If a Voltage Source of n no. ~~of~~ with Voltages V_1, V_2, V_3 upto V_n with $R_1, R_2 \ldots$ upto R_n (Internal resistances) are connected in parallel then these Voltage Sources can be replaced by a Single Voltage Source. (V) in Series with a resistance (R).



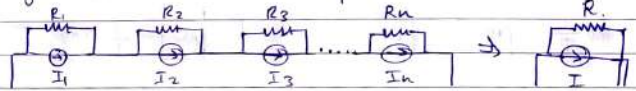
$V_1, V_2 \ldots V_n$; $R_1, R_2 \ldots R_n$; $I = I_1 + I_2 + I_3 \cdots + I_n$

$$\frac{V}{R_{eq}} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \cdots \frac{V_n}{R_n}$$

$$R_{eq} = \frac{1}{R_1} + \frac{1}{R_2} \cdots \frac{1}{R_n}$$

$$V = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \cdots \frac{V_n}{R_n}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \cdots \frac{1}{R_n}}$$

(ii) If n no. of current Source with current $I_1, I_2, I_3 \cdots I_n$ are connected in Series with internal resistance $R_1, R_2, R_3 \ldots R_n$ are connected in parallel with the current Source. Then the Current Source can be replaced by a Single Current Source I in parallel with a resistance R.



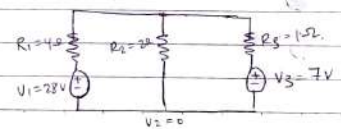
$$I = \frac{V}{R}$$

$V = V_1 + V_2 + V_3 \cdots + V_n \Rightarrow V = I_1R_1 + I_2R_2 + \cdots + I_nR_n$.

$R = R_1 + R_2 + R_3 \cdots + R_n$

$$I = \frac{I_1R_1 + I_2R_2 \cdots + I_nR_n}{R_1 + R_2 \cdots + R_n}$$

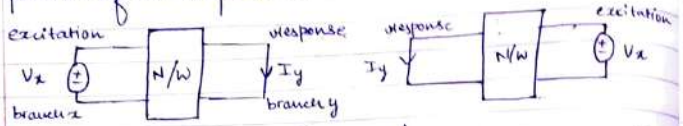
Q. Find the Voltage across all the branches in the below given circuit by Millmann's Theorem.



$$V = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{1/R_1 + 1/R_2 + 1/R_3} = \frac{\frac{28}{4} + \frac{0}{2} + \frac{7}{1}}{1/4 + 1/2 + 1/1} = \underline{8V}$$

* Reciprocity Theorem

In a linear single source network with ratio of response to excitation remains the same even when the position of the response and excitation are changed.



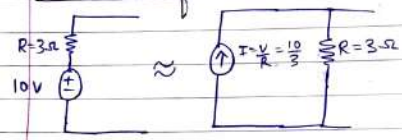
$$\left.\frac{\text{Response}}{\text{Excitation}}\right|_1 = \left.\frac{\text{Response}}{\text{Excitation}}\right|_2$$

Q. Find the value of current I when the network satisfy the Reciprocity theorem.

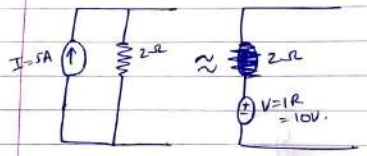
$$\frac{2}{10} = \frac{I}{30}$$

$\underline{I = 6A}$

* Source Transformation:



Voltage to current

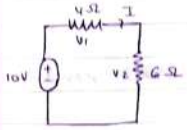


Current to Voltage

* Substitution Theorem

Any branch in a network can be substituted by a different branch without disturbing the voltages and current of the entire network provided the new branch has the same terminal voltage as the original branch.

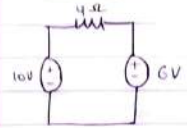
e.g.



$I = V/R = 10/6+4 = \frac{10}{10} = \underline{1A}$

$V_1 = I \times R_1 = 1 \times 4 = \underline{4V}$

$V_2 = I \times R_2 = 1 \times 6 = \underline{6V}$



$V = 10 - 6 = \underline{4V}$

$I = \frac{4}{4} = \underline{1A}$

→ Peak factor / Crest factor / Amplitude factor of a Periodic wave = Ratio of maximum value to the RMS value.

$$\text{Peak factor} = \frac{I_{max}}{I_{RMS}} = \frac{I_{max}}{I_{max}} \times \sqrt{2} = \underline{1.414}$$

Q. Write the mathematical expression for a 50 Hz sinusoidal voltage supplied for a domestic purpose at 230V.

$\omega = 2\pi f$

$= 2 \times \pi \times 50 \Rightarrow 100\pi = \underline{314}$

$V_m = 230 \times \sqrt{2} = 325.22. \Rightarrow V = 325.22 \sin 314t$

27/9/24

Q. The equation of an alternating current $i = 141.4 \sin 314t$ what is the RMS value and frequency.

$\omega = 2\pi f$

$314 = 2 \times 3.14 f$

$f = \underline{50\ Hz.}$

$I_{rms} = \frac{I_{max}}{\sqrt{2}} = \frac{141.4}{1.414} = \underline{100A}$

Q. An AC frequency 50Hz and RMS 25A. Write the equation of the current and find:

i) Current at time 0.0025 sec.

ii) Time at which current is 1.414 A.

$I_{max} = 25 \times \sqrt{2} = \underline{35.35\ A.}$

$\omega = 2 \times 3.14 \times 50 = \underline{314}$

(1) $I = I_{max} \sin \omega t$

$I = 35.35 \sin 314 \times 0.0025$

$I = 35.35 \sin 0.785 = \underline{24.9A}$

(2) $1.414 = 35.35 \sin 314t$

$314t = \frac{1.414}{35.35} \sin^{-1} \Rightarrow t = \frac{1}{314} \sin^{-1} 0.04$

→ Advantages of AC over DC:

1. More voltage can be generated (upto 33,000 V) and DC (upto 650V).
2. AC voltage can be increased and decreased with the help of a static machine called transformer.
3. AC transmission and distribution is more economical as line material (copper) can be saved by transmitting power at higher voltage.
4. AC motors for the same horse power as of DC motors are cheaper, lighter in weight required less space and required low maintainance.
5. AC can be converted into DC easily when and where required but DC cannot be converted to A.C easily.

→ Applications of DC:

1. DC motors are used in railways, cranes and lifts.
2. For electroplating, electrolytic and electro-chemical process, DC is required.
3. Search lights and cinema projectors work on DC.
4. Relay and operating time switches etc. and circuit breakers, DC works more efficiently.
5. In paper mills DC motors are required.